ARNAV MANKAD

ENGINEERING PROJECT PORTFOLIO

Biomedical engineer combining hands-on medical device development, experimental R&D;, computational modeling, signal processing, machine learning, and medical imaging.

01	**Soft Optical Sensor for Bleeding Detection**	Medical device R&D
02	**Continuous-Flow Bioreactor**	Fluidics & modeling
03	**Foundation Models for Medical Image Segmentation**	Deep learning & imaging
04	**Morphological Profiling for CRISPR Target Discovery**	Computational biology

Johns Hopkins University | M.S.E. Biomedical Engineering - Biomedical Data Science, Expected May 2027
Boston University | B.S. Biomedical Engineering, Data Science Minor - Summa Cum Laude

amankad2@jh.edu

PROJECT 01

Soft Optical Sensor for Bleeding Detection

Boston University Material Robotics Laboratory | Medical device research for colonoscopy

Medical Devices	Sensors	Fluidics	MATLAB	IMU	Prototyping

THE CHALLENGE

Develop a soft sensing approach capable of detecting accidental bleeding during colonoscopy while operating in a fluid-rich, highly constrained environment.

MY CONTRIBUTIONS

- Fabricated soft optical sensors using soft-lithography techniques and became the primary researcher responsible for sensor fabrication.
- Engineered a fluidic filtering approach to reduce interference from unwanted bodily fluids entering the sensor microchannels.
- Developed a MATLAB interface for real-time blood-loss information and contributed to machine-learning approaches for probabilistic bleeding detection.
- Developed an IMU-based localization approach using orientation tracking and Kalman filtering to help map the sensor's path and localize detected bleeding sites.
- Supported the transition toward a wireless sensor, including PCB design and soldering.
- Participated in ex vivo evaluation at Brigham and Women's Hospital.

OUTCOME

The project progressed from soft-sensor fabrication and benchtop development to ex vivo testing, wireless integration, and localization. The work contributed to two peer-reviewed publications in Device (Cell Press/Elsevier) and Advanced Sensor Research (Wiley).

ENGINEERING TAKEAWAY

This project taught me how to iterate a biomedical device across fabrication, fluidics, sensing, electronics, algorithms, and experimental validation rather than treating each discipline in isolation.

PROJECT 02

Continuous-Flow Bioreactor

Boston University Senior Design | Reproducing physiological hormone fluctuations in vitro

CAD	COMSOL	Fluidics	Testing	Assays	System Design

THE CHALLENGE

Design a benchtop system capable of delivering controlled, time-varying hormone concentrations to an in vitro culture environment instead of relying on static exposure conditions.

MY CONTRIBUTIONS

- Designed and iterated bioreactor and fluidic components using CAD in Onshape.
- Used COMSOL fluid-flow and mixing simulations to evaluate design behavior before physical testing.
- Helped integrate programmable pumping to control fluid delivery and generate changing concentration profiles.
- Conducted benchtop experiments using a fluorescence-based assay to quantify mixing performance.
- Used simulation and experimental results together to identify design changes and improve system behavior.

DESIGN APPROACH

The project followed an iterative model-build-test workflow: computational models were used to understand mixing and flow, physical prototypes were constructed, and experimental measurements were used to evaluate whether the system behaved as intended.

ENGINEERING TAKEAWAY

The project strengthened my experience translating a biological requirement into a physical system and using both simulation and experiments to guide engineering decisions.

PROJECT 03

Foundation Models for Medical Image Segmentation

Johns Hopkins University | Low-label 3D head & neck CT segmentation

Python	Deep Learning	3D CT	nnU-Net	VISTA3D	Experiment Design

RESEARCH QUESTION

Can a pretrained medical-imaging foundation model reduce the amount of expert annotation needed for accurate 3D CT segmentation compared with training a task-specific model from scratch?

EXPERIMENTAL DESIGN

- Compare nnU-Net trained from scratch with the pretrained NV-Segment-CT/VISTA3D model on the SegRap2023 dataset.
- Evaluate performance across progressively larger labeled-data budgets using the same cases for both model families.
- Study both fully labeled volumes and sparse/partial annotation conditions.
- Compare fine-tuning strategies, including frozen and more extensive fine-tuning configurations.
- Evaluate segmentation performance using Dice as the primary metric, with sensitivity, boundary metrics, variability, and computational requirements as complementary measures.

WHY IT MATTERS

High-quality 3D medical-image annotations are expensive and time-consuming to produce. If pretrained models retain strong performance with substantially fewer labels, they could make segmentation systems easier to develop for new clinical tasks and anatomies.

ENGINEERING TAKEAWAY

This project combines model implementation with careful experimental design: controlling data splits, label budgets, evaluation metrics, and fine-tuning choices so that model comparisons are meaningful.

PROJECT 04

Morphological Profiling for CRISPR Target Discovery

Computational biology project | Discovering structure in high-dimensional Cell Painting data

Python	Machine Learning	Fuzzy K-Means	Cell Painting	Data Analysis

THE CHALLENGE

Use high-dimensional morphological measurements from Cell Painting experiments to identify latent patterns that could help characterize gene perturbations and support CRISPR target discovery.

MY CONTRIBUTIONS

- Developed the analysis approach and Python implementation for fuzzy k-means clustering of high-dimensional morphological profiles.
- Represented samples through probabilistic membership across latent morphological states rather than forcing each observation into a single hard cluster.
- Aggregated and analyzed gene-level latent-state distributions to compare morphological behavior across perturbations.
- Explored similarities in morphological profiles as a way to organize and interpret complex biological screening data.

WHY FUZZY CLUSTERING?

Biological phenotypes do not always fall into clean, mutually exclusive groups. Fuzzy clustering allows a sample to express partial membership in multiple latent states, preserving ambiguity that may be biologically meaningful.

ENGINEERING TAKEAWAY

The project expanded my experience applying machine-learning methods to noisy, high-dimensional biomedical data and translating an algorithmic representation into an interpretable biological analysis.

TECHNICAL TOOLKIT

A multidisciplinary toolkit spanning physical prototyping, computational modeling, data science, and biomedical research.

Programming & Data	Python, MATLAB, machine learning, data analysis, signal processing, deep learning
Design & Simulation	SolidWorks, Onshape, COMSOL, engineering design, fluidics, prototyping
Medical Device R&D;	Sensor development, soft lithography, experimental testing, ex vivo evaluation, PCB assembly
Algorithms	Kalman filtering, localization, computer vision, medical image segmentation, clustering
Workflow	Jupyter, VS Code, GitHub, Docker, technical documentation, interdisciplinary collaboration

ABOUT

I am a biomedical engineer interested in medical device R&D;, product development, systems engineering, biomedical data science, and health technology. I enjoy projects where I can move between hands-on engineering and computation - building, testing, analyzing, and iterating until a system works better.

Arnav Mankad | amankad2@jh.edu

Johns Hopkins University - M.S.E. Biomedical Engineering, Biomedical Data Science